

Important Notice Before Using the MentorPi Robot

The MentorPi robot is based on ROS and involves multiple modules and relatively complex operations. To ensure you have the best experience using this product, we strongly recommend reading this document in advance.

This guide is intended for users with a basic background in electronics, information technology, and computer science. Familiarity with Linux, Python, and ROS is recommended.

To support your learning, we have included several foundational tutorials in the resource package. You can find them in the “1 Tutorial Materials” folder, including:

- Chapter 2: Linux Basic Lesson
- Chapter 3: Python Basic Lesson
- Chapter 4: OpenCV Computer Vision Course
- Chapter 5: ROS2 Basic Course

1. Important Notes Before Use

1.1 Battery Charging

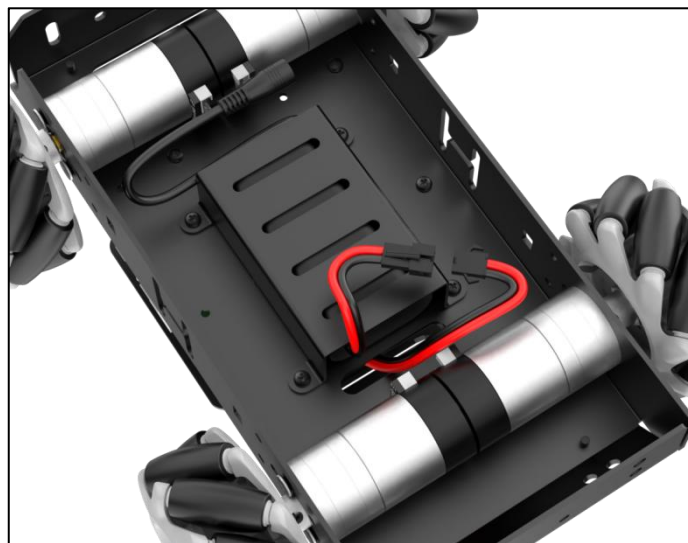
Caution:

1. Do not connect the battery lead wires directly to the lithium battery before connecting it to the controller. Direct contact of the positive and negative terminals may cause a short circuit.
 2. To ensure stable operation of the robot, please charge the battery promptly when the voltage drops below 6.4V. Use only the charger provided in the kit for charging.
-

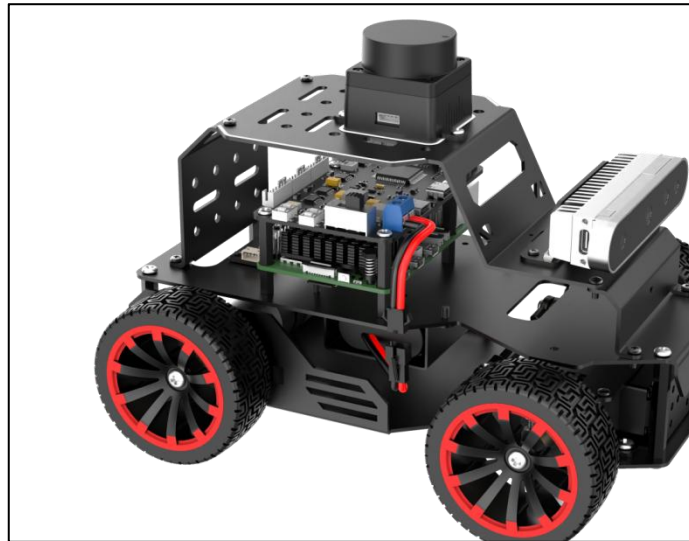
Due to transportation regulations, lithium batteries are not fully charged when shipped. Before first use, please fully charge the battery. Users should first connect the battery to the lead wires, then proceed with charging. Charging typically takes around 1 hour.

Charging Steps:

- 1) Make sure the device is turned off.
- 2) Match the battery wires by color: connect red to red and black to black.

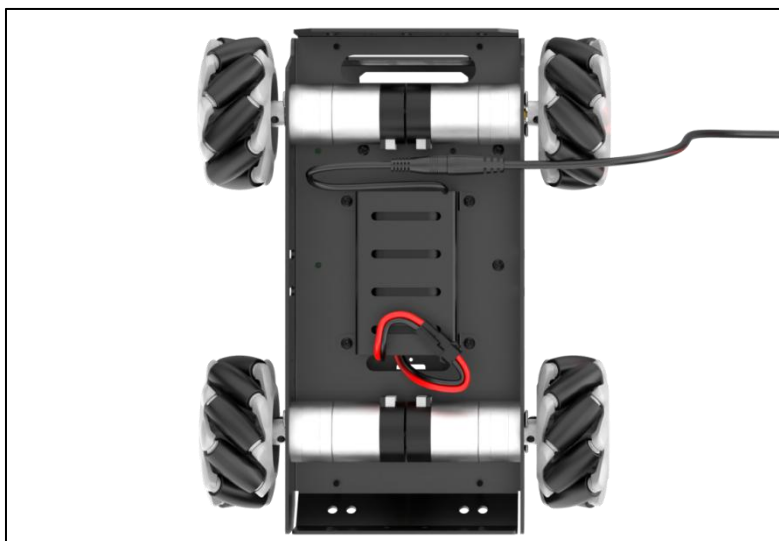


Wiring Diagram for Mecanum Chassis

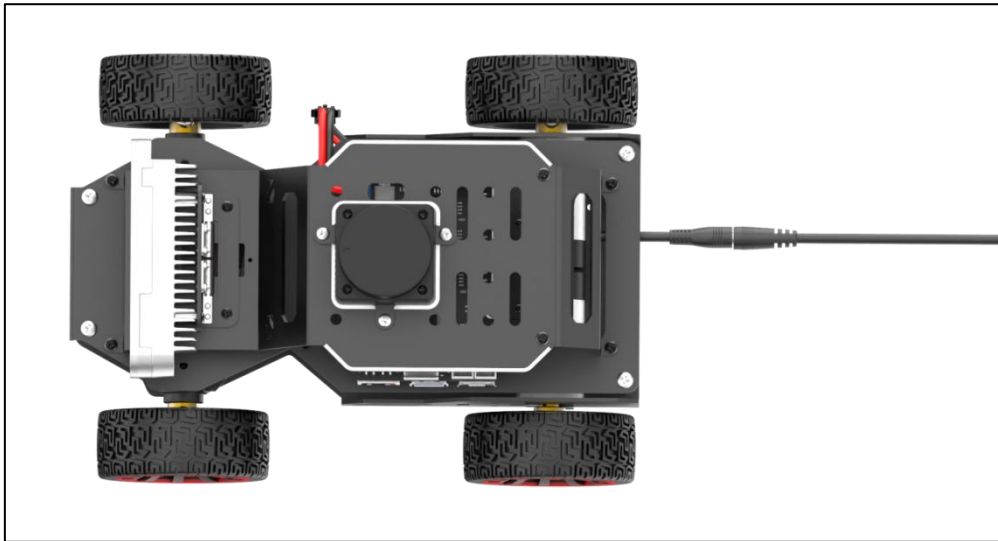


Wiring Diagram for Ackermann Chassis

- 3) Remove the lithium battery's DC charging port from the side or bottom of the robot, then insert the charger plug to begin charging.



Charging Port Location on Mecanum Chassis



Charging Port Location on Ackermann Chassis

- 4) You can check the charging status via the indicator light on the charger:



Reminder: Once charging is complete, please unplug the charger promptly to prevent overcharging.

1.2 Robot Assembly

The assembly of the MentorPi robot involves numerous detailed steps, so it is not fully covered in this document. For a comprehensive guide, please refer to the folder:

“1. Getting Ready\ 1.2 Quick Start Guide\ 1.2.2 Assembly”

1.3 System Configuration After Power-On

The MentorPi robot supports different chassis types (Mecanum wheel, Ackermann) and camera versions. If you need to switch between different robot configurations or camera models, please follow the steps below:

Note:

- If the chassis type is incorrectly configured, the robot may not respond to motion commands properly, or its movement trajectory and direction may behave abnormally.
- If the wrong camera type is selected, image data may not appear on the mobile app or Raspberry Pi desktop interface.

In such cases, please refer to the following instructions to reconfigure the system.

- 1) Power on the robot and connect it to the remote control software **VNC**.

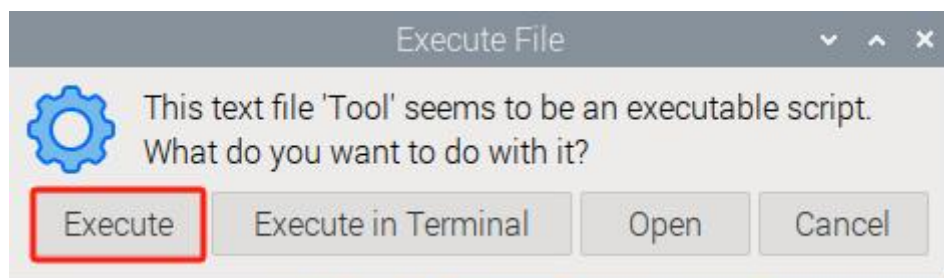
For guidance on connecting via remote desktop, refer to: **8. Set**

Development Environment\ Lesson 1 VNC Installation and

Connection

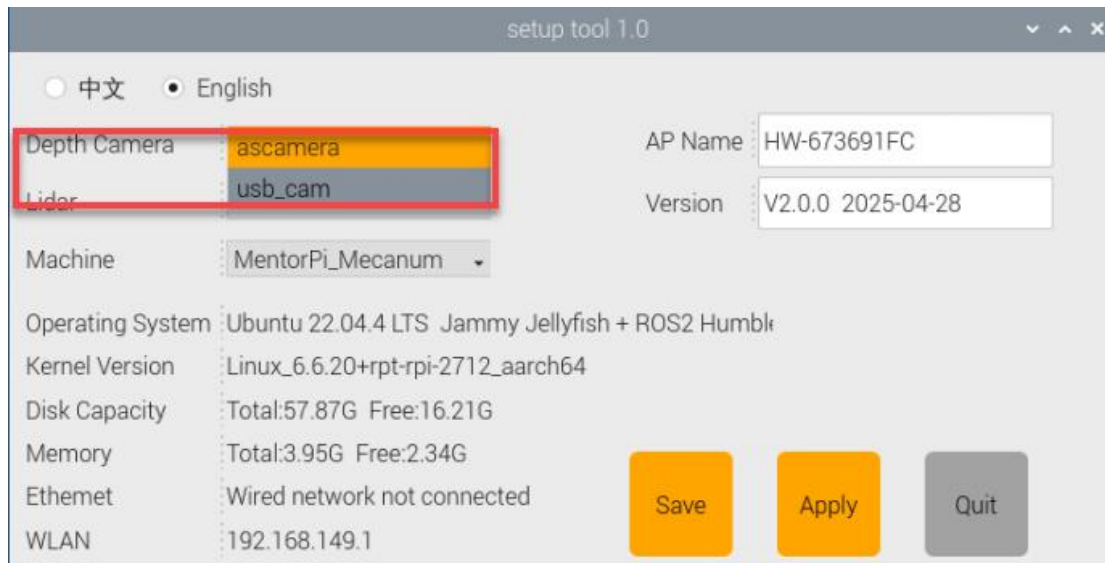


- 2) On the remote desktop, double-click the “**Tool**” icon to launch the system configuration utility, then click the “**Execute**” button.



- 3) In the interface, under **Camera Type**, select the appropriate camera

version:



- ① **ascamera**: Orbbec depth camera
- ② **usb_cam**: Standard monocular camera with 2D pan-tilt

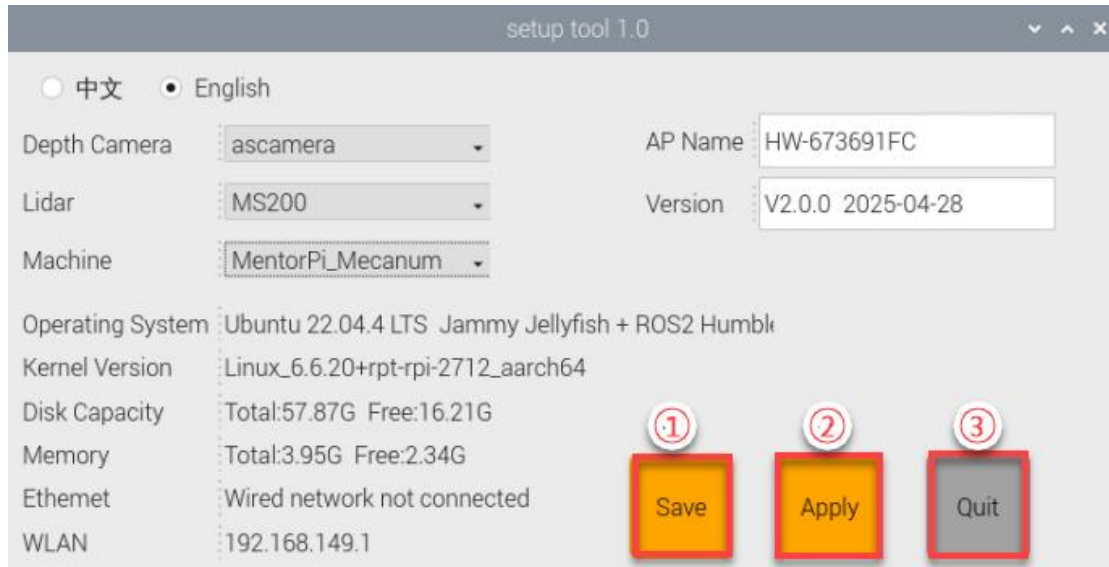
4) Under Robot Type, select the corresponding chassis:



- ① **MentorPi_Mecanum**: Mecanum wheel chassis
- ② **MentorPi_Acker**: Ackermann chassis

5) After making your selections, click **Save** → **Apply** → **Exit** in order. It's

crucial to follow this sequence to ensure the settings take effect correctly.



- 6) Wait for the buzzer to emit a short beep, indicating that the chassis type has been successfully switched. You can open the terminal to check the ROS 2 environment configuration .
- 7) Once the configuration is complete, proceed to: 1. Getting Ready\ 1.4 App Control; 1.5 Wireless Controller Control.